

Waste-to-energy: Biogas potential of waste from coffee production and consumption

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panel [Wojciech Czekata](#)^a, Aleksandra Łukomska^a, Jakub Pulka^a, Wiktor Bojarski^a, Patrycja Pochwatka^b, [Alina Kowalczyk-Juśko](#)^b, [Anna Oniszczyk](#)^c, [Jacek Dach](#)^a

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Highlights

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Biogas efficiency of coffee husk was $329.50 \text{ m}^3 \cdot \text{Mg}^{-1}$ fresh matter.

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Biogas efficiency of spent coffee grounds was in range of $225.45\text{--}270.97 \text{ m}^3 \cdot \text{Mg}^{-1}$ FM.

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Both substrates were characterized by similar methane concentrations $52.68\text{--}54.50\%$.

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Coffee waste (especially coffee grounds) has short HRT favorable for biogas plants.

Abstract

The generation of waste is a necessary process related to the production and consumption of coffee. One of the wastes associated with the production of coffee is the husk, which separates from the beans when it is roasted. The waste from the consumption of coffee is spent coffee grounds generated during the coffee brewing process. The study aimed to determine the possibility of coffee production and consumption waste conversion in the [anaerobic digestion](#) process. The tested parameters were, among others, [biogas](#) efficiency, methane efficiency, and methane content in [biogas](#). The experiment was conducted under [anaerobic digestion](#) conditions using DIN 38414/S8 norm. The total solids content of the coffee husk was 93.37%, with the organic matter content at 93.34%. The biogas efficiency in terms of fresh matter was $329.50 \text{ m}^3 \cdot \text{Mg}^{-1}$. The total solids content of all three analyzed grounds samples ranged from 41.27 to 45.72%.

The high volatile solids content, in the 97.91–98.41% range, confirmed the biogas potential. In all three samples of coffee grounds, biogas efficiency was in the range of 225.45–270.97 m³·Mg⁻¹. The obtained results allow concluding that analyzed coffee production and consumption waste have a high potential for biogas production.

Graphical abstract



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Introduction

Coffee is a beverage made from coffee beans obtained from perennial and tropical plants. Beans are morphologically variable and have different shapes, colors, and sizes. Internally, seeds (usually two per fruit) are found, processed, and used to prepare infusions [1]. The most popular types of coffee are Arabica and Robusta [2], which are grown in regions stretching around the equator and the areas between the Tropic of Cancer and Capricorn [3]. The choice of the place of cultivation largely depends on the type of coffee and the climatic and soil requirements. The seeds are planted in shaded beds or containers filled with organically enriched soil. After sprouting, the seedling is replanted into separate pots, and then, after reaching sufficient dimensions, it can be planted in the ground. The fruiting time for the coffee tree is about 20 years [4]. Unprocessed coffee beans are green and yellow-orange shortly before maturation. When they are ready for harvest, they turn bright red. The coffee fruit is divided into three layers: epicarp or skin, which is the outermost layer; mesocarp or pulp, which forms a sweet and aromatic pulp of mucilaginous nature, protected by a cellulose layer called parchment or endocarp; and finally, a silvery layer, which covers the two oval-shaped grains called endosperm. The leading coffee producers are in South American countries. The highest consumption occurs in Europe, although this beverage is drunk regularly worldwide [5].

The generation of waste is a necessary process related to the production and consumption of coffee [6]. The processing of the coffee beans causes the layers to be lost and to reach the beans. Only 5% of the bean is used to produce a coffee crop, and the rest remains in a residual form as husk leaves, branches, green fruits, pulp, mucilage, parchment, and silverskin. On the other hand, spent coffee grounds are waste generated during the coffee brewing process [7]. Spent coffee grounds contain lipids, carbohydrates, and nitrogen-containing compounds [8]. The increase in coffee consumption in the world is the reason for producing more and more waste and by-products. For each ton of coffee consumed, around 650 kg of residue is generated. In 2020–2021, world coffee consumption was almost 10 million tons, meaning about 6.5 million tons of coffee grounds were produced [9]. Analogously to other waste, spent coffee grounds should be managed according to the law [10].

The basic directions of waste management related to the production and consumption of coffee are their storage. Landfilling of biodegradable waste is not the preferred option and causes emissions that can harm the environment [11]. One of the most popular alternatives is using them for fertilization, e.g., in gardening. During the decomposition of the coffee grounds, humic substances are produced. These compounds are responsible for the development of the plant, its metabolism, and the hormone balance. Also, reducing the C:N from 26:1 to about 10:1. Adding coffee grounds to fertilizer can bind pesticide residues and toxic heavy metals, and use as mulch reduces soil temperature and increases soil moisture content. In addition, coffee grounds can serve as food for bio-consumers, such as earthworms, which aerate the soil and improve the nutritional supply of plants [12]. The use of coffee grounds for fertilization has a positive effect on the development of crops, however, it does not solve the problem of their emissivity.

Moreover, using them in this way does not fulfill their potential. A much better way is to use coffee grounds as additives, e.g., in cosmetics, food, and clothing products. Particularly noteworthy, in the context of the ongoing energy transformation, is the possibility of using coffee grounds to produce energy. Most of the waste associated with the production and consumption of coffee consists of an organic substance and are classified as biomass [13]. They can be transformed into thermal and biological processes. In the case of waste with a high total solids content, it is possible to use them to produce solid biofuels [14]. This group includes, among others, coffee husk [15]. Lignin, which in this process is responsible for binding structures in the biomass, accounts for approximately 33% of their volume [16]. It is also possible to burn them directly or blend them with other waste without prior processing. Another solution is the pyrolysis process. Among others, the possibility of processing coffee husk in the discussed process was examined by Dal-Bó et

al. [17]. Waste containing more water should be dried before it is transferred to incineration processes. However, this generates additional costs.

Biological processing processes are an alternative to thermal processes [18]. This group includes composting and anaerobic digestion [19,20]. Composting of coffee waste has been widely discussed in the literature. Liu and Price [21] analyzed composting systems to manage spent coffee grounds. The conversion of spent coffee grounds into vermicompost was presented, among others, in research by Hanc et al. [22].

On the other hand, Sołowiej et al. [23] analyzed the effect of heat removal during the thermophilic phase on the energetic aspects of the biowaste composting process with the addition of spent coffee grounds. Considering the properties of waste and by-products related to the production of coffee and the growing energy demand, methods of energy use of the discussed substrates are becoming increasingly popular [24]. Apart from the products mentioned above of solid biofuels, including briquettes and pellets, attention should be paid to other processes [25]. For example, spent coffee grounds can be used to produce liquid biofuels. Coffee oil must be extracted to use coffee or waste related to its production. Among others, research on using spent coffee grounds for biodiesel production was presented by Caetano et al. [26]. On the other hand, the study by Kwon et al. [27] raised the issue of the co-production of biodiesel and bioethanol with spent coffee grounds.

The direction of using high and low-water-content waste is anaerobic digestion. As a result of the anaerobic digestion process, biogas is generated from substrates rich in an organic matter [28]. Its main component is methane. It should be emphasized that waste and by-products related to the production and consumption of coffee can be used to produce gaseous biofuels. Coffee grounds, mainly due to the content of compounds such as carbohydrates, proteins, or fats, are perfect as a substrate for a biogas plant. As previously mentioned, lipids, a component of coffee grounds, can significantly affect the amount of biogas produced. Due to the fragmentation, the discussed biomass does not require pre-treatment, which brings financial benefits and shortens the time it takes to prepare it for the process. It is also essential that the grounds can be easily separated from the rest of the waste immediately after the coffee brewing process. It means that such a substrate usually does not contain foreign materials, and thus the fermentation process itself is easier to monitor. In addition to the high content of lignin and other organic substances that positively affect the process, the grounds contain an appropriate C:N ratio, which was previously necessary when used as a fertilizer [29].

Another advantage of coffee grounds is the fact that despite the presence of potentially harmful compounds such as caffeine or polyphenols, they do not inhibit the process. The

produced biogas is classified as a renewable energy source [30]. For this reason, the technology of methane production should be considered for many wastes from the agri-food and food sectors [31], especially in the context of systematically rising natural gas prices. Additionally, the digestate produced can be successfully used in fertilization [32]. It can allow reducing the demand for mineral fertilizers.

With the growing energy demand, alternative energy sources are required. Due to issues related to the circular economy, the preferred solution is to use waste and by-products for energy production. For this reason, research on waste related to coffee production and consumption was carried out. The study aimed to determine the possibility of coffee production and consumption waste conversion in the anaerobic digestion process. The tested parameters were, among others, biogas efficiency, methane efficiency, and methane concentration in biogas. According to the authors, the prepared article can fill some knowledge gaps and indicate further action directions.

Materials

The substrate used for the research was the coffee husk obtained from the coffee bean processing company. Additionally, spent coffee grounds obtained from three cafes in Poznań (Poland) were used in the research. The substrates used in the research were analyzed to determine the basic physical and chemical parameters. Total solids were measured using the drying method (24 h at 105 °C) according to the Polish Standard PN-75 C-04616/01, volatile solids by burning dried samples at 525 °C for 3 h

Substrate characteristics

The total solids of coffee husk were 93.37%, and the content of volatile solids was 93.34% of the total solids. The total solid of the spent coffee grounds ranged from 41.27 to 45.72% for all three samples. Volatile solids for these three samples ranged from 97.91 for sample C to 98.42% for sample B. The pH for all analyzed substrates was similar and ranged from 5.23 to 5.48 (Table 1).

Methane fermentation tests

The decomposition time of the substrates in the fermentation reactor varies. This parameter depends on the

Conclusions and further research directions

The production and consumption of coffee generate a significant amount of waste that should be rationally managed. It would help avoid or reduce the negative impact on the environment. One of the wastes associated with the production of coffee is the husk, which separates from the beans when it is roasted. The waste from the consumption of

coffee is spent coffee grounds generated during the coffee brewing process. Due to their properties, both of the wastes mentioned can be used, among others,

Credit author statement

Wojciech Czekala: Conceptualization, Methodology, Validation, Investigation, Resources, Writing - Original Draft, Writing - Review & Editing, Visualization, Supervision, Funding acquisition. **Aleksandra Łukomska:** Conceptualization, Methodology, Investigation, Resources, Writing - Original Draft, Writing - Review & Editing, **Jakub Pulka:** Formal analysis, Writing - Original Draft, Writing - Review & Editing, **Wiktor Bojarski:** Writing - Original Draft, Writing - Review & Editing, **Patrycja Pochwatka:**

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The publication was co-financed within the framework of the Ministry of Science and Higher Education program as “Regional Initiative Excellence” in the years 2019–2022, Project No. 005/RID/2018/19.

The paper’s authors would like to thank the people who helped obtain the material for the research and the laboratory worker.